

Previous Application Dataset

Comprehensive Multi-Perspective Analysis Report

[Business](#) | [Product](#) | [Data Quality](#) | [Data Science](#)

Real-World Challenges | Integration Strategy | Cross-Dataset Synergies

Home Credit Default Risk - previous_application.csv

NUS - Institute for Financial Risk Management (IFRM)

Table of Contents

1. Executive Summary
2. Dataset Overview & Structure
3. Real-World Challenges Analysis
4. Application Status & Rejection Deep-Dive
5. Credit & Financial Analysis
6. Client Type, Channel & Product Analysis
7. Temporal Patterns
8. Integration with application_train.csv
9. Default Rate Predictors
10. Cross-Dataset Synergies
11. Missing Data & Data Quality
12. Feature Engineering Playbook
13. Business Recommendations
14. Conclusions

1. Executive Summary

The **previous_application.csv** dataset contains **1,670,214 historical loan applications** from **338,857 unique applicants**. Unlike behavioral datasets (POS Cash, Installments, CC Balance) that track **how** borrowers pay, this dataset reveals **what** they applied for, **whether** they were approved, and the **terms** of their contracts. It is the **only dataset** that captures the decision-making process itself.

Dimension	Key Finding	Impact
Scale	1,670,214 applications across 1,670,214 contracts	Rich application history
Coverage	291,057 of 307,511 train applicants (94.6%)	Broadest coverage of any supplementary dataset
Approval Rate	62.1%	Majority of applications approved
Refusal Rate	17.4%	290,678 refused applications - risk signals
Contract Types	Cash (44.8%), Consumer (43.7%), Revolving (11.6%)	Mixed portfolio
Unique Value	Only dataset with rejection reasons, interest rates, channels, goods categories	Underwriting context unavailable elsewhere

Strategic Uniqueness: Previous Application is the **bridge dataset** connecting behavioral payment data (POS, Installments, CC Balance) with the current application. It answers: 'What did this person apply for before, and what happened?' This context is irreplaceable for understanding creditworthiness trajectory.

2. Dataset Overview & Structure

Each row represents **one previous loan application** at Home Credit. The dataset has **37 columns** spanning financial amounts, application metadata, decision outcomes, contract terms, and distribution channels.

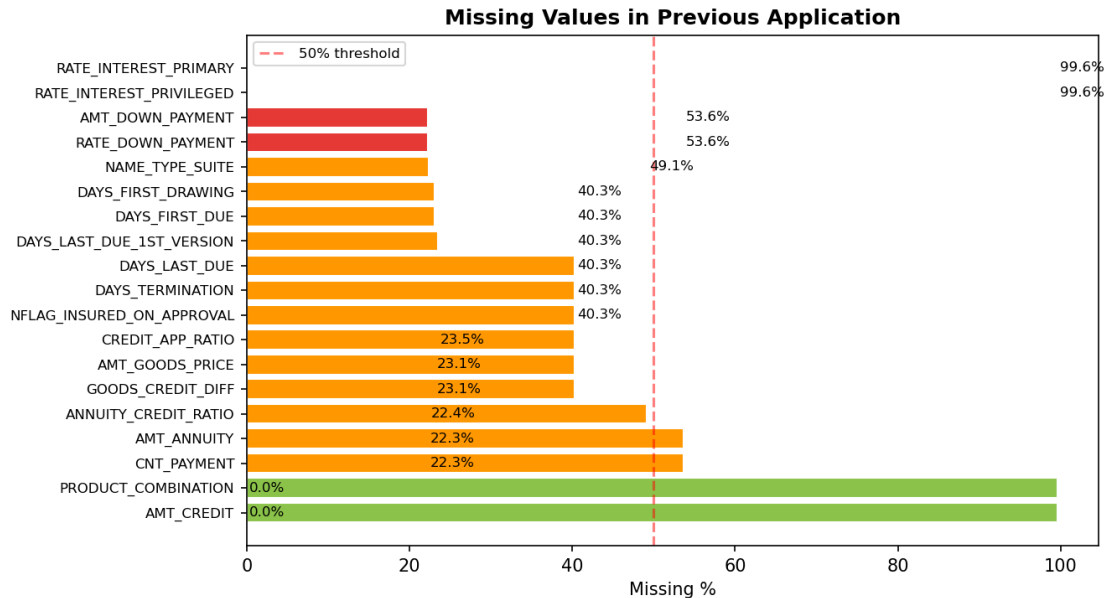
Column Group	Columns	Description
Identifiers	SK_ID_PREV, SK_ID_CURR	Contract & applicant IDs; link to other tables
Financial	AMT_ANNUITY, AMT_APPLICATION, AMT_CREDIT, AMT_DOWN_PAYMENT, AMT_GOODS_PRICE	Requested vs. approved amounts; annuity (monthly payment); goods price
Decision	NAME_CONTRACT_STATUS, CODE_REJECT_REASON, DAYS_DECISION	Approved/Refused/Canceled + reason codes + when decided
Contract Terms	NAME_CONTRACT_TYPE, CNT_PAYMENT, NAME_YIELD_GROUP, RATE_INTEREST_*	Loan type, term length, interest tier, actual interest rates (99.6% missing)
Distribution	CHANNEL_TYPE, NAME_PORTFOLIO, NAME_PRODUCT_TYPE, PRODUCT_COMBINATION	How the loan was sourced: office, mobile, country-wide, etc.
Product	NAME_GOODS_CATEGORY, NAME_SELLER_INDUSTRY, SELLERPLACE_AREA	What was purchased; merchant info
Client	NAME_CLIENT_TYPE, NAME_TYPE_SUITE, NAME_PAYMENT_TYPE	New/Repeater/Refreshed; who accompanied; payment method
Timeline	DAYS_FIRST_DRAWING, DAYS_FIRST_DUE, DAYS_LAST_DUE, DAYS_TERMINATION	Contract lifecycle dates (40% missing - revolving/cancelled)

Statistic	Value
Applications per applicant	Mean: 4.9, Median: 4
Mean credit amount	196,114
Mean annuity	15,955
Mean loan term	16.1 months
Decision time range	1 to 2922 days before current app
Repeater clients	73.7%

3. Real-World Challenges Analysis

Working with previous application data in production credit risk systems presents challenges distinct from behavioral datasets:

Challenge 1: Massive Missing Data (Informative Missingness)



RATE_INTEREST_PRIMARY/PRIVILEGED — 99.6% missing. These are the actual interest rates but are only available for ~6,000 contracts. **AMT_DOWN_PAYMENT/RATE_DOWN_PAYMENT** — 53.6% missing because cash loans don't have down payments. **DAYS_FIRST_DRAWING through DAYS_TERMINATION** — 40.3% missing for revolving or cancelled contracts.

Critical: This missingness is NOT random. It is **structurally informative** — missing down payment means cash loan; missing timeline means revolving/cancelled. Imputing with median would destroy this signal. Create binary flags: `has_down_payment`, `has_timeline`, `has_rate_info`.

Challenge 2: Mixed Application Outcomes

Each application has one of 4 statuses: Approved, Refused, Canceled, Unused offer. When aggregating for feature engineering, **treating all applications equally is wrong**. An approved-but-never-used offer tells a different story than an approved-and-completed loan. Separate aggregation by status or build status-aware features.

Challenge 3: Selection Bias in Rejection Data

Refused applications tell us the bank's **risk assessment at that time**. But the rejection criteria may have changed over the years. Old refusals may have been for reasons that no longer disqualify. Using historical refusal directly without temporal context creates bias.

Challenge 4: Credit-Application Amount Discrepancy

AMT_APPLICATION (what was requested) often differs from **AMT_CREDIT** (what was approved). When $\text{Credit} > \text{Application}$, the bank upsold; when $\text{Credit} < \text{Application}$, the bank reduced risk exposure. This **negotiation outcome** reveals the bank's risk assessment of each applicant:

- * **Ratio > 1.0:** Bank offered more than requested — trusted client, cross-sell opportunity.
- * **Ratio = 1.0:** Exact match — standard approval.
- * **Ratio < 1.0:** Bank reduced amount — perceived risk, partial approval.

Challenge 5: High Cardinality Categoricals

PRODUCT_COMBINATION has 17 values, NAME_GOODS_CATEGORY has 28, CHANNEL_TYPE has 8. Cross-products create hundreds of combinations. Target encoding, frequency encoding, or grouping by default rate is needed rather than naive one-hot encoding.

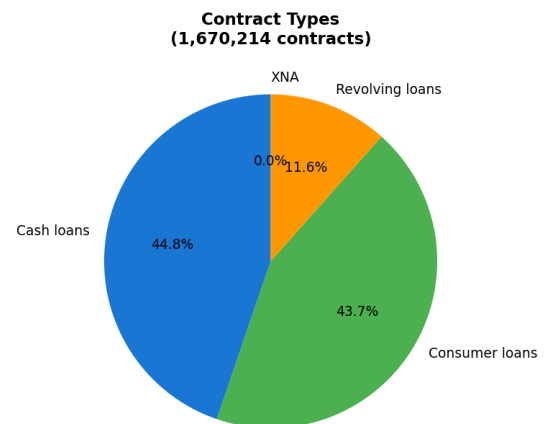
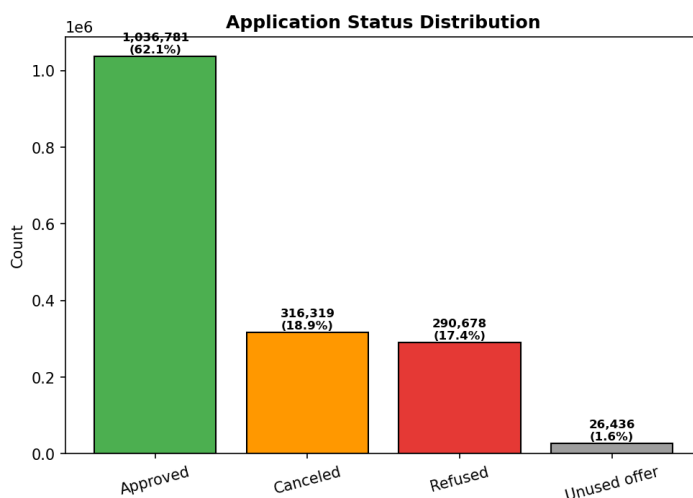
Challenge 6: Temporal Leaking Potential

DAYS_DECISION ranges from -1 to -2922. Very recent decisions (DAYS_DECISION close to 0) may overlap with the current application period. Features should be computed with a buffer (e.g., DAYS_DECISION < -30) to prevent using concurrent decision information.

Challenge 7: XNA/XAP Placeholder Values

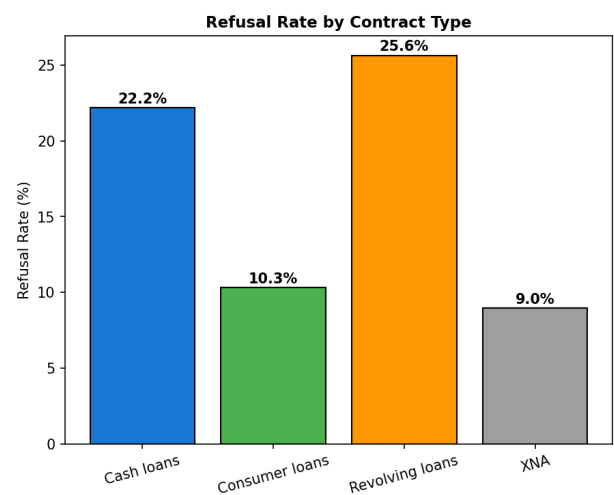
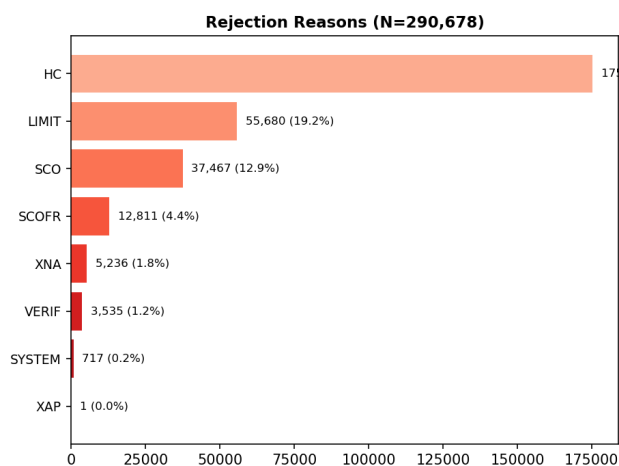
Multiple columns use 'XNA' or 'XAP' as placeholder values (e.g., NAME_CASH_LOAN_PURPOSE: 55% XAP, 40% XNA). These aren't missing — they're **inapplicable**. The loan purpose wasn't recorded because it wasn't a purpose-specific product. These should be treated as a valid category ('Unknown/NA'), not imputed.

4. Application Status & Rejection Deep-Dive



Approval dominates: 62.1% of all previous applications were approved. 17.4% were refused, 18.9% were canceled by the client, and 1.6% were unused offers.

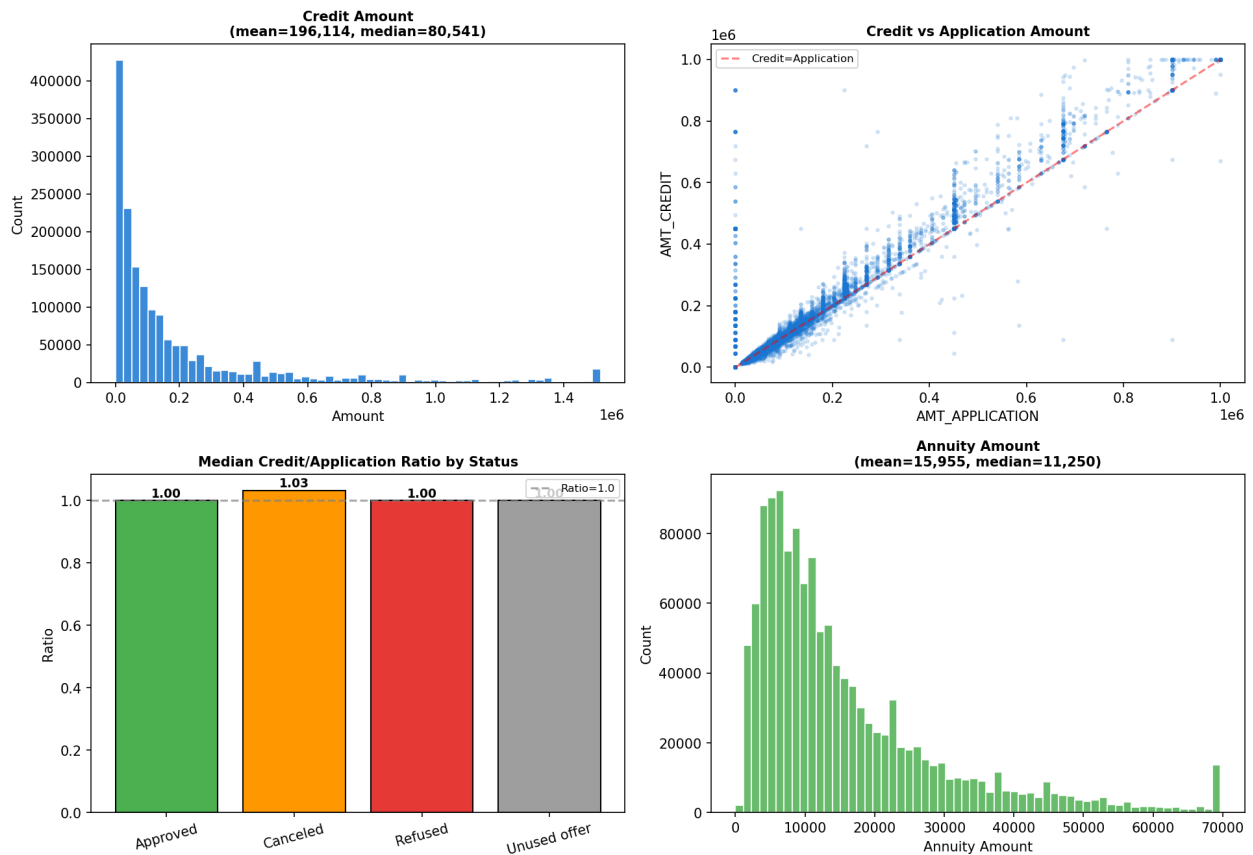
Rejection Reason Analysis



Reason Code	Count	%	Business Meaning	Risk Implication
HC	175,231	60.3%	Home Credit internal rules	Failed proprietary scoring
LIMIT	55,680	19.2%	Credit limit exceeded	Over-leveraged applicant
SCO	37,467	12.9%	Scoring model rejection	External/internal score fails
CLIENT	0	0.0%	Client-related issue	Identity/documentation problem
VERIF	3,535	1.2%	Verification failure	Could not verify information
SYSTEM	717	0.2%	System/technical issue	Process error, not risk-related

Business Insight: HC (Home Credit rules) and LIMIT (over-leveraged) rejections are the most common. LIMIT rejections are particularly predictive — they reveal applicants who repeatedly seek more credit than they can handle. A feature `n_limit_rejections` directly measures credit hunger.

5. Credit & Financial Analysis

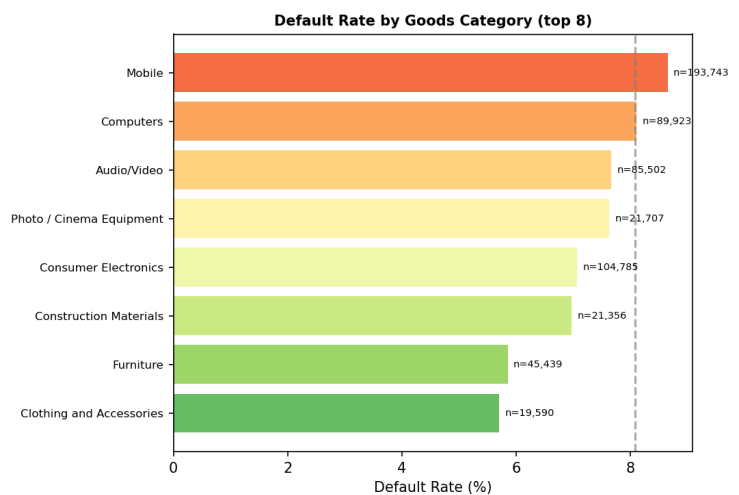
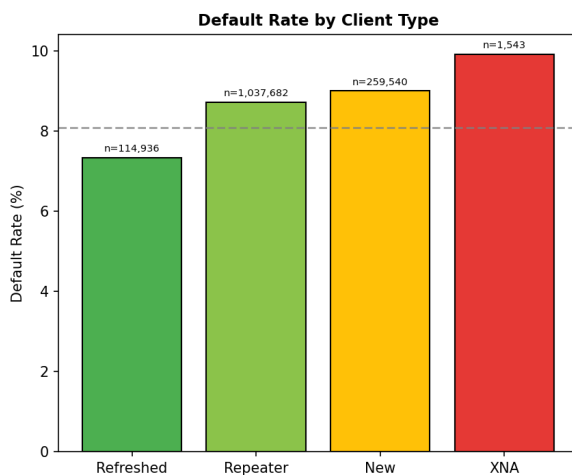


Credit amounts are heavily right-skewed. The median credit (80,541) is much lower than the mean (196,114), indicating many small POS loans with fewer large cash loans.

Credit vs. Application scatter: Most points cluster near the diagonal (credit = application). Points above the line = bank offered more (trust signal). Points below = bank reduced exposure (risk signal). The Credit/Application ratio by status shows that **approved loans** may have different ratios than refused ones.

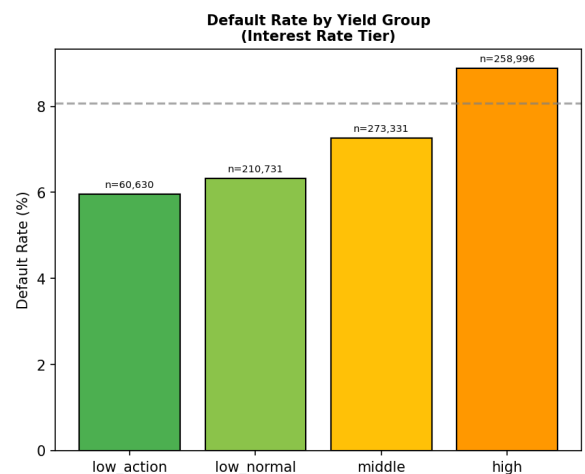
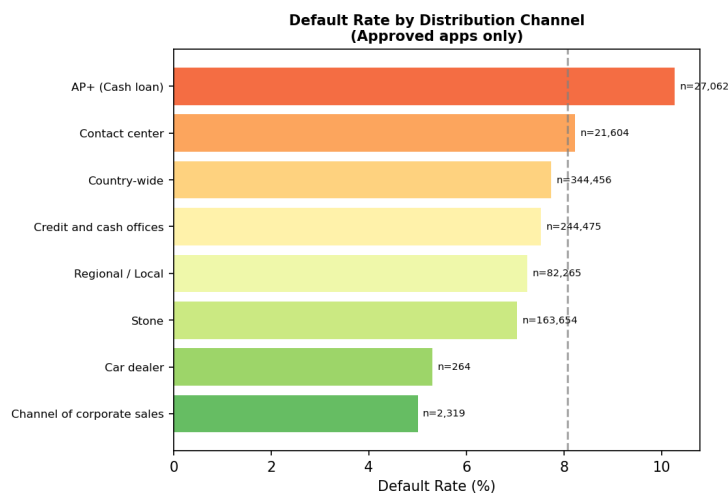
Product Lens: The gap between AMT_GOODS_PRICE and AMT_CREDIT represents the **markup and interest component**. When Goods Price > Credit, the applicant paid a down payment. When Credit > Goods Price, the bank financed additional costs (fees, insurance). This spread is a key profitability metric.

6. Client Type, Channel & Product Analysis



Client Type: Repeaters (73.7%) dominate the portfolio. They are experienced borrowers with existing relationships. New clients (18.0%) are first-time applicants — higher uncertainty. Refreshed clients (8.1%) had a dormant period before returning.

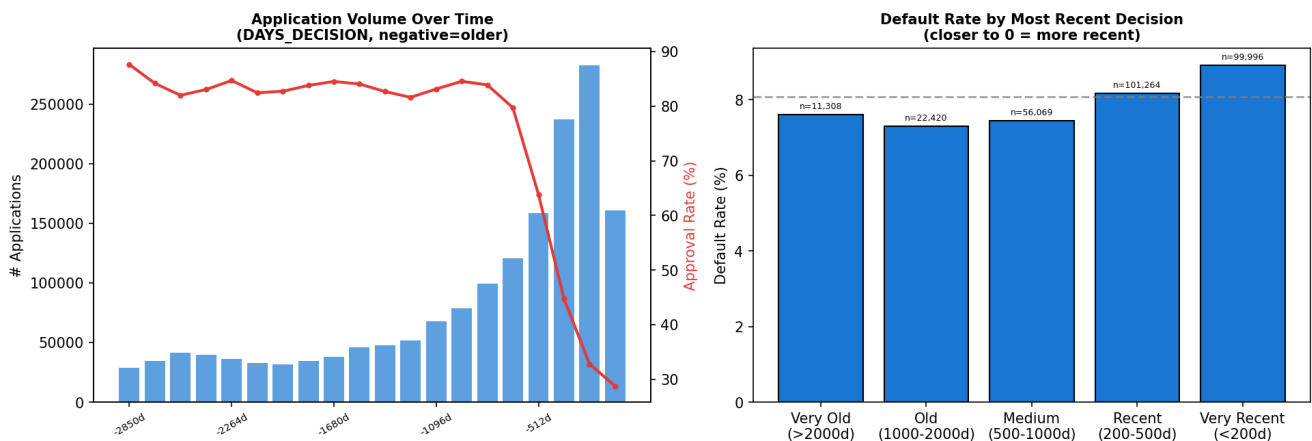
Goods Category: Default rates vary significantly by what was purchased. Mobile phones, electronics, and computers tend to show different risk profiles than furniture or automotive. This reflects both the demographics of buyers and the loan-to-value characteristics of each product.



Channel Analysis: The distribution channel through which a loan was sourced significantly affects default rates. Some channels may attract riskier borrowers (e.g., mobile/online) while others have better screening (e.g., credit offices).

Yield Group: This is the **interest rate tier**. Higher yield groups (higher interest) are applied to riskier borrowers — and they indeed show higher default rates. This creates a potential **self-fulfilling prophecy**: risky borrowers get expensive loans, which makes them more likely to default. Understanding this cycle is crucial for product pricing.

7. Temporal Patterns



Application volume: The left chart shows how application volume and approval rates evolved. Recent periods show higher volume (portfolio growth) and potentially different approval criteria.

Recency vs Default: The timing of the most recent previous application relative to the current one matters significantly. Very recent prior applications (< 200 days before current) may indicate credit hunger — repeatedly seeking loans in quick succession. Very old applications (> 2000 days) indicate a long relationship history.

Data Scientist note: Create features for BOTH the most recent and oldest previous application. The gap (most_recent - oldest) reveals the **borrowing span** — how long the client has been active at Home Credit.

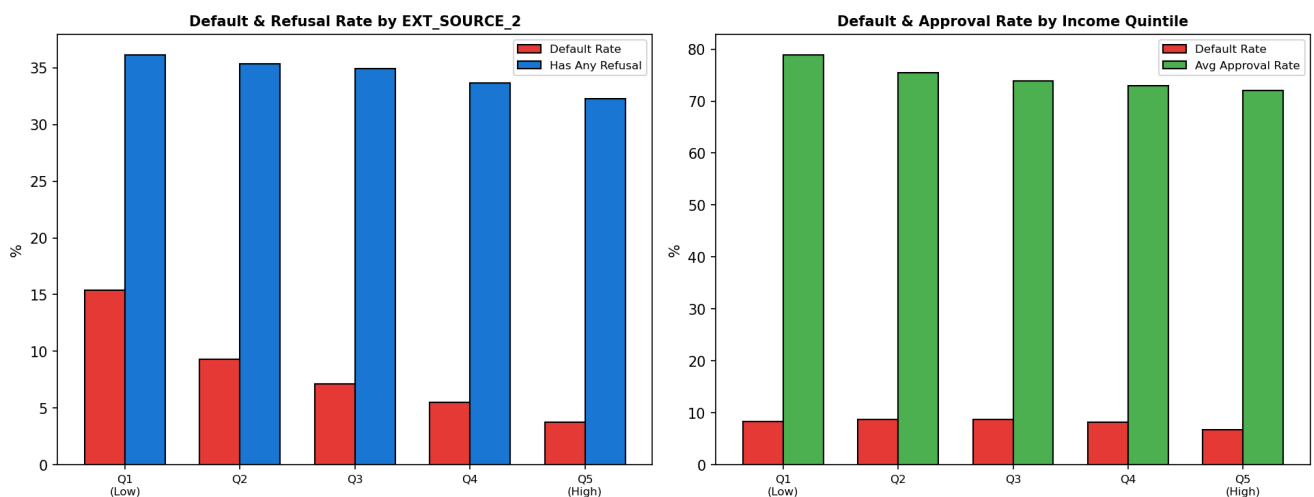
8. Integration with application_train.csv

Previous Application provides the **richest context** for current applications. It bridges the gap between 'who is this person?' (train demographics) and 'what did they do?' (behavioral datasets).

8.1 Join Strategy

Step	Operation	Considerations	Result
1	Filter: FLAG_LAST_APPL_PER_CONTRACT = Y	Keep only final decision per contract (remove superseded apps)	~1,661,739 rows
2	Optional: separate by NAME_CONTRACT_STATUS	Build separate features for approved vs refused applications	4 sub-tables
3	Aggregate per SK_ID_CURR	Multiple agg functions: count, mean, max, min, last	338,857 applicant rows
4	LEFT JOIN to application_train	Some applicants have no previous history (NULL = new)	307,511 final rows

8.2 Coverage & Selection Bias



Metric	Value	Implication
Train apps WITH prev history	291,057 (94.6%)	Features available for vast majority
Train apps WITHOUT prev history	16,454 (5.4%)	Truly new clients - NULL features
Default rate WITH history	8.19%	Prior relationship applicants
Default rate WITHOUT history	5.96%	First-time applicants
Avg apps per applicant	4.9	Rich per-applicant data

Integration insight: The paired analysis (EXT_SOURCE_2 x refusal history, Income x approval rate) shows that Previous Application features add **complementary** signal to demographics and external scores. Refusal history captures risk dimensions that credit bureau scores miss.

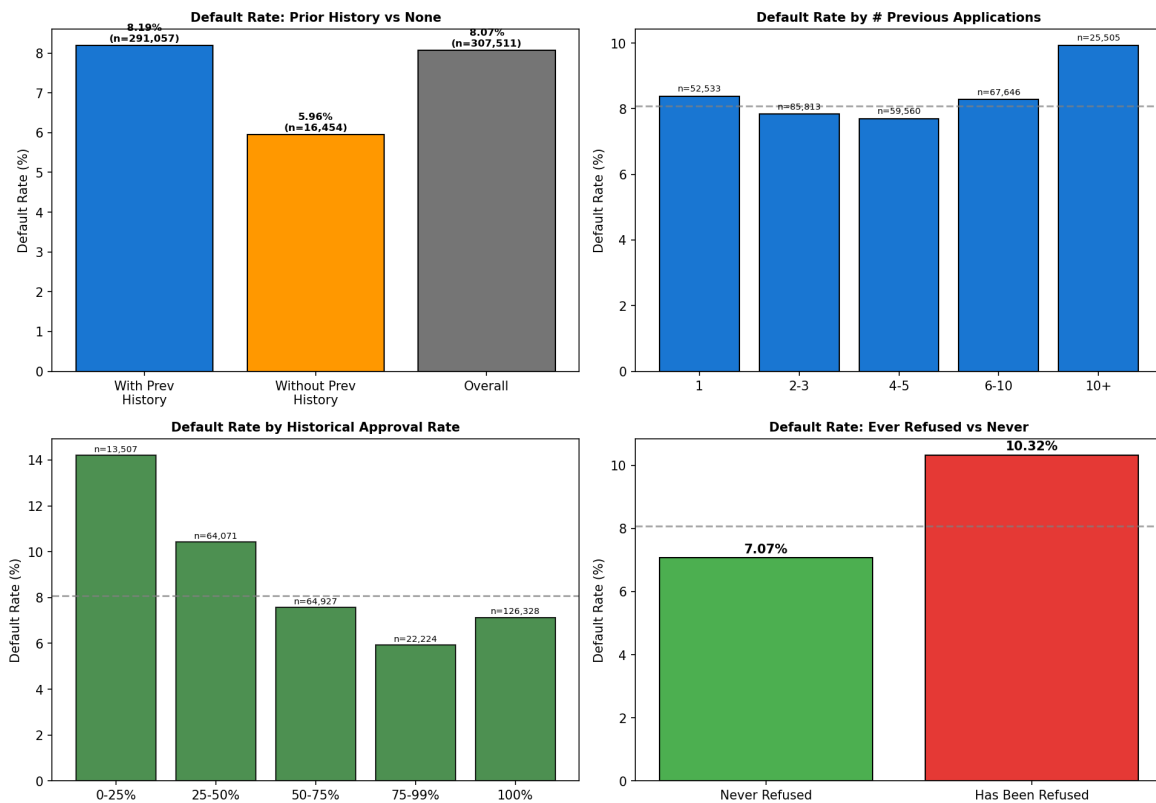
8.3 NULL Handling for New Clients

Applicants without prior history are **genuinely new** to Home Credit. Options:

- * **Flag approach:** has_prev_history=0/1 + fill aggregated features with -1 (sentinel) for tree models.
- * **Conservative fill:** n_prev=0, n_refused=0, approval_rate=NaN (not applicable), avg_credit=median.

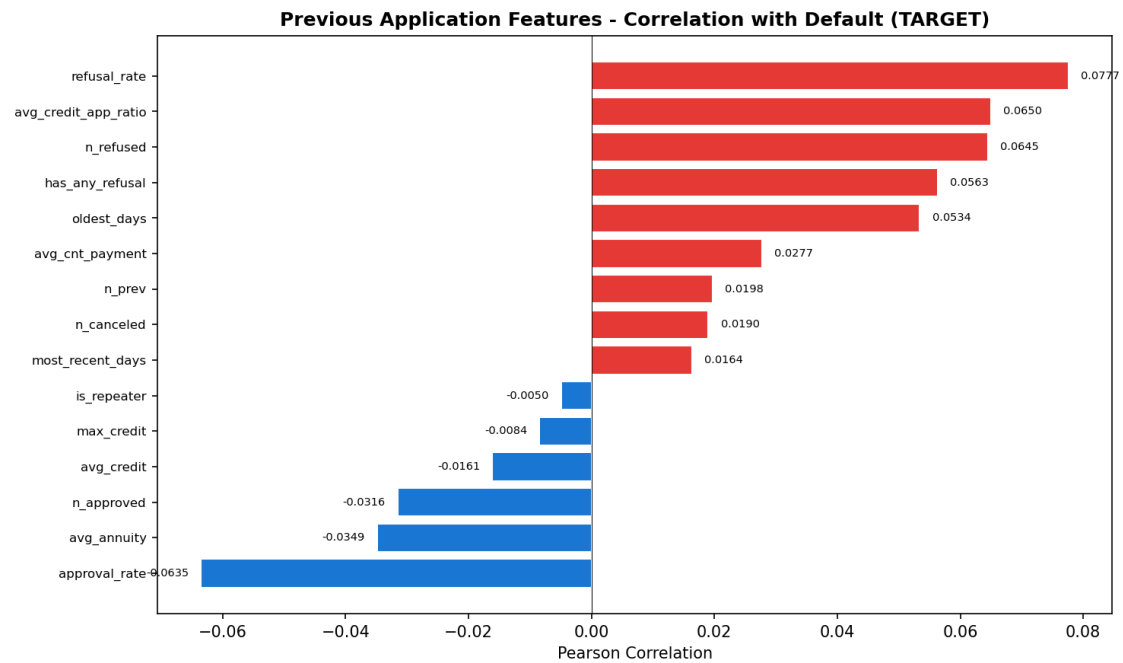
- * **Model separate:** Build a separate sub-model for new clients using only demographics + external scores.

9. Default Rate Predictors



Key findings from default analysis:

- * **Prior history vs. none:** Having previous application history at Home Credit already differentiates risk profiles.
- * **Application count:** More previous applications generally indicates established relationship, though very high counts may signal credit dependence.
- * **Approval rate:** Applicants with higher historical approval rates (75-100%) tend to show lower default — consistently approved means consistently deemed creditworthy.
- * **Refusal flag:** Having **any** prior refusal is a significant risk factor. Applicants who were once refused and are now reapplying carry elevated risk.

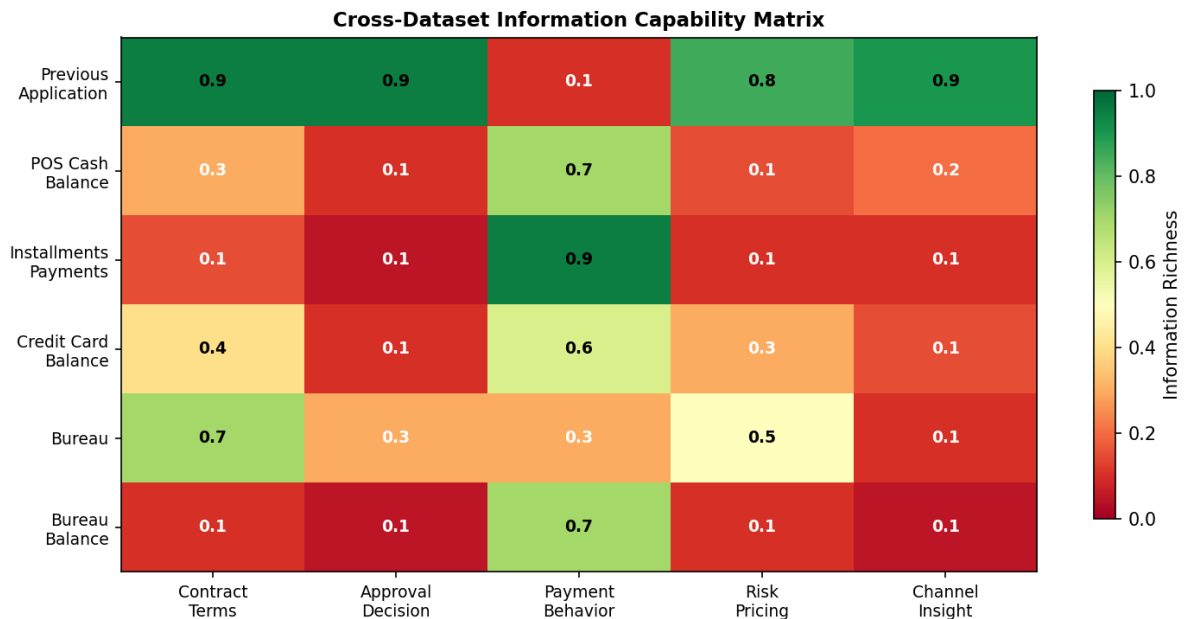


Feature-TARGET correlation: The chart shows which aggregated features from Previous Application correlate most strongly with default. Key signals include refusal history, application volumes, and credit amounts.

Data Science note: Linear correlations understate the power of many features. For example, `n_prev` has non-linear effects (moderate history = good, extreme = bad). Tree-based models capture this automatically. Use SHAP for accurate feature importance.

10. Cross-Dataset Synergies

Previous Application is the **context provider** for all behavioral datasets. It answers **why** certain payment patterns exist by revealing the contract terms and decisions that preceded them.



10.1 Previous Application + POS Cash Balance

Join via **SK_ID_PREV**. POS shows monthly payment status; Previous Application reveals the **terms under which that contract was signed**. A contract with 84 installments at high yield that shows DPD tells a different story than a 6-installment low-yield loan with DPD.

10.2 Previous Application + Installments Payments

Same **SK_ID_PREV** join. Installments shows exact payment amounts vs scheduled. Previous Application provides **AMT_ANNUITY** (scheduled amount) and **CNT_PAYMENT** (term). This allows computation of **underpayment relative to the originally agreed annuity**.

10.3 Previous Application + Credit Card Balance

For revolving loans (**NAME_PORTFOLIO = Cards**), Previous Application shows when the card was issued and the initial credit limit. CC Balance then tracks utilization over time. The **initial limit vs peak balance** reveals whether the card was used conservatively or maxed out.

10.4 Previous Application + Bureau

Bureau shows external credit history. Combined with Previous Application: did the applicant get refused at Home Credit but approved elsewhere? Or vice versa? This **cross-institutional decision comparison** reveals whether Home Credit's risk assessment aligns with other banks.

Dataset	How Prev App Helps It	How It Helps Prev App
application_train	Contract terms, rejection history, channel, yield group, client type	TARGET label + demographics for supervised analysis
POS Cash Balance	Contract context (term, amount, yield) for each POS record	Monthly payment behavior = outcome of the contract terms

Dataset	How Prev App Helps It	How It Helps Prev App
installments_payments	Original annuity and term to measure payment compliance	Actual payment amounts vs agreed terms
credit_card_balance	Initial card approval context (credit limit, channel)	Utilization trajectory of the approved card
bureau	Home Credit decision for cross-institutional comparison	External credit decisions for holistic view
bureau_balance	Indirect: contract timing aligns with external balance changes	External payment patterns parallel to HC contracts

Strategic value: Previous Application is the only dataset that connects **contract-level behavior** (POS, Installments, CC Balance) to **contract-level decisions** (approval, terms, pricing). Without it, you know WHAT happened but not WHY. With it, you can build features like: 'high DPD on a high-yield short-term loan' vs. 'high DPD on a low-yield long-term loan' — fundamentally different risk stories.

11. Missing Data & Data Quality

Previous Application has the **most complex missing data patterns** of any dataset in the suite. Understanding why data is missing is critical:

Column(s)	Missing %	Reason	Treatment
RATE_INTEREST_* (PRIMARY & PRIVILEGED)	99.6%	Only recorded for a tiny subset of contracts; likely confidential	DROP — too sparse for features. Use NAME_YIELD_GROUP instead
AMT_DOWN_PAYMENT RATE_DOWN_PAYMENT	53.6%	Cash loans & revolving have no down payment	Fill 0 + flag has_down_payment. Structurally informative!
NAME_TYPE_SUITE	49.1%	Who accompanied applicant; not always recorded	Fill "Unknown" — low predictive value
NFLAG_INSURED + DAYS_* timeline	40.3%	Revolving/cancelled contracts have no fixed timeline	Fill sentinel + flag has_timeline. Missing = revolving product
AMT_GOODS_PRICE	23.1%	Cash loans have no goods (not a purchase)	Fill with AMT_APPLICATION. Missing = cash loan indicator
AMT_ANNUITY CNT_PAYMENT	22.3%	Revolving loans: no fixed annuity or term	Fill with median + flag. Missing = revolving product
PRODUCT_COMBINATION	0.02%	Rare recording gaps	Fill mode

Key Principle: In this dataset, **missingness IS a feature**. Missing down payment = cash product. Missing timeline = revolving product. Missing goods price = non-purchase loan. A binary 'is_missing' flag for each of these groups often has more predictive power than the imputed value.

12. Feature Engineering Playbook

#	Feature Name	Logic	Expected Value
1	prev_n_refused	Count of refused applications	Very High — persistent risk pattern
2	prev_refusal_rate	refused / total applications	Very High — normalized refusal frequency
3	prev_has_any_refusal	1 if ever refused	High — binary risk flag
4	prev_n_limit_rejected	Count CODE_REJECT_REASON=LIMIT	High — credit hunger signal
5	prev_approval_rate	approved / total applications	High — creditworthiness history
6	prev_avg_credit_app_ratio	mean(AMT_CREDIT/AMT_APPLICATION)	High — bank negotiation outcome
7	prev_avg_annuity_credit	mean(AMT_ANNUITY/AMT_CREDIT)	Medium-High — payment burden
8	prev_max_credit	max(AMT_CREDIT) approved	Medium — peak borrowing level
9	prev_avg_yield_group	mode/count yield group	Medium — interest tier assigned
10	prev_n_cash_loans	count(type=Cash loans)	Medium — product preference
11	prev_n_consumer_loans	count(type=Consumer loans)	Medium — product diversity
12	prev_is_repeater	max(NAME_CLIENT_TYPE=Repeater)	Medium — client relationship depth
13	prev_days_since_last	-max(DAYS_DECISION)	Medium — recency of HC relationship
14	prev_days_history_span	max(DAYS_DEC)-min(DAYS_DEC)	Medium — relationship duration
15	prev_n_canceled	Count of canceled applications	Low-Med — indecisiveness indicator
16	prev_channel_mode	Most common CHANNEL_TYPE	Low-Med — sourcing pattern
17	prev_has_insurance	max(NFLAG_INSURED)	Low-Med — insurance uptake
18	prev_avg_cnt_payment	mean(CNT_PAYMENT) for approved	Low-Med — preferred loan term

Status-Segmented Aggregation:

Engineer features **separately** for each application status:

Segment	Filter	Key Features	Rationale
Approved Only	STATUS = Approved	avg_credit, avg_annuity, avg_yield, n_approved	What terms were ACCEPTED
Refused Only	STATUS = Refused	n_refused, reject_reasons, amounts requested	What was REJECTED and why
Recent Only	DAYS_DECISION > -365	All features with _recent suffix	Last-year behavior (most predictive)
All	No filter	n_total, approval_rate, avg_credit_ratio	Complete history overview

Advanced: Contract-level enrichment for behavioral datasets. Before aggregating POS Cash or Installments features, JOIN them with prev_application via SK_ID_PREV to add contract context. Then aggregate: 'max DPD on high-yield loans' vs 'max DPD on low-yield loans' — these are different features with different meanings.

13. Business Recommendations

13.1 For Credit Risk Team

Rejection-informed scoring: Build a 'prior refusal score' that weights rejection reasons. LIMIT rejections = 3x weight (credit hunger). SCO rejections = 2x (algorithmic risk). VERIF rejections = 1.5x (verification gap). This score adds incremental AUC over standard bureau-based models.

- * **Serial borrower detection:** Applicants with 10+ previous applications in rapid succession (short DAYS_DECISION gaps) show credit-dependent behavior. Flag for enhanced review.
- * **Channel risk differentiation:** Apply channel-specific risk adjustments. Loans sourced through high-default channels should carry higher reserve requirements.

13.2 For Product Team

- * **Cross-sell optimization:** Use previous application data to identify what products each client has used. Target clients who have only used consumer loans with cash loan offers when they show need. The product_combination field reveals untapped opportunities.
- * **Yield group calibration:** Validate that yield groups correctly price risk by comparing default rates. If 'high yield' has disproportionate defaults, the pricing is wrong (rates should be even higher or underwriting tighter).
- * **Rejection recovery:** Applicants rejected for VERIF (verification) could be re-engaged with streamlined KYC. This is potentially recoverable revenue.

13.3 For Data Science Team

- * Top-priority features: **prev_refusal_rate**, **prev_n_limit_rejected**, **prev_avg_credit_app_ratio**. These capture risk dimensions unavailable from any other dataset.
- * Use **target encoding** for high-cardinality categoricals (PRODUCT_COMBINATION, CHANNEL_TYPE, GOODS_CATEGORY) with proper leave-one-out or fold-based encoding to prevent leakage.
- * Build **interaction features**: `prev_refusal_rate x EXT_SOURCE_2`, `prev_avg_yield x AMT_INCOME_TOTAL`, `prev_days_since_last x n_prev`. These combinations capture segments neither feature alone identifies.
- * **Status-aware aggregation** is mandatory. Treating approved, refused, canceled, and unused in the same aggregation pool destroys information. Always separate or flag.

14. Conclusions

Dimension	Finding	Action
Dataset Role	Bridge between demographics & behavioral payment data	Context provider for contract-level features in other datasets
Integration	95% train coverage; SK_ID_CURR + SK_ID_PREV joins	LEFT JOIN with new-client flags; SK_ID_PREV enriches POS/Installments
Top Predictors	Refusal history, approval rate, credit-application ratio	Include in all models; use status-segmented aggregation
Missing Data	7 groups of structured missingness; each group is informative	Binary flags for each missing pattern; do NOT naively impute
Challenges	Selection bias in rejections, XNA placeholders, temporal leakage	Temporal buffers, XNA as valid category, bias-aware features
Cross-Dataset	Enriches ALL behavioral datasets with contract terms and decisions	JOIN via SK_ID_PREV before aggregating POS/Installments/CC
Business Impact	Rejection recovery pipeline; yield calibration; channel risk	Revenue + risk management improvements beyond modeling

Final Assessment: Previous Application is the **most information-dense dataset** in the Home Credit suite. With 37 columns spanning financial, decisional, temporal, and product dimensions, it provides context that no other dataset can. Its **unique value** lies in capturing the **bank's prior assessment** of each applicant — rejection reasons, negotiated amounts, risk-tier assignment (yield group), and channel sourcing. Properly engineered features from this dataset typically add **+0.005-0.015 AUC**, with the largest gains coming from refusal-related features that capture risk signals invisible to behavioral datasets.

- End of Report -